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Validating urban scaling laws through mobile phone data: analysis of Brazil's largest cities

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Abstract

Urban scaling theories posit that larger cities exhibit disproportionately higher levels of socioeconomic activity and human interactions. Yet, evidence from developing countries (especially those marked by stark socioeconomic disparities) remains limited. To address this gap, we analyse a month-long dataset of 3.1 billion voice-call records from Brazil's 100 most populous cities, providing a national-scale test of urban scaling laws. We measure interactions using two complementary proxies: the number of phone-based contacts (voice-call degrees) and the number of trips inferred from consecutive calls in distinct locations. Our findings reveal clear superlinear relationships in both metrics, indicating that larger urban centres exhibit intensified remote communication and physical mobility. We further observe that gross domestic product (GDP) also scales superlinearly with population, consistent with broader claims that economic output grows faster than city size. Conversely, the number of antennas required per user scales sublinearly, suggesting economies of scale in telecommunications infrastructure. Although the dataset covers a single provider, its widespread coverage in major cities supports the robustness of the results. We nonetheless discuss potential biases, including city-specific marketing campaigns and predominantly prepaid users, as well as the open question of whether higher interaction drives wealth or vice versa. Overall, this study enriches our understanding of urban scaling especially in the global south, emphasising how communication and mobility jointly shape the socioeconomic landscapes of rapidly growing cities.

Keywords Urban scaling, CDR data, Human mobility, Social interactions

1 Introduction

Researchers from various disciplines have identified consistent empirical relationships between city population size and a wide range of urban metrics. These findings point to a remarkable regularity: as cities grow, many of their characteristics change in systematic and predictable ways. For instance, as city population increases, there is a corresponding rise in the per capita number of patents, average salaries, and other socioeconomic indicators (Bettencourt et al., 2007), the so-called *increasing return to scale* in larger cities. Conversely, as city population increases, there is a corresponding decrease in the per capita number of petrol stations, built-up area, and other infrastructure-related variables (Kühnert et al., 2006;

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Ribeiro et al., 2025). This pervasive pattern, observed across diverse urban systems, is captured by the concept of *urban scaling laws*, statistical regularities that describe how urban quantities scale with population size (Pumain et al., 2006; Bettencourt et al., 2007; Ribeiro and Rybski, 2023). Although the mechanisms underlying these relationships are debated within urban science (Arcaute et al., 2015; Keuschnigg et al., 2019; Florida, 2003), and the validity of these regularities remains under discussion (Naroll and von Bertalanffy, 1973; Ribeiro and Rybski, 2023; Lobo et al., 2020), urban scaling laws have gained increasing prominence through integrative theoretical frameworks drawing from biology, physics, and economics (Naroll and von Bertalanffy, 1973; Barenblatt, 1996; West, 2017; Marquet et al., 2005; Keuschnigg, 2019).

Urban scaling laws provide a robust framework for examining how population size influences socioeconomic activities, knowledge production, and infrastructure needs, thereby shaping the social fabric of cities (Arbesman et al., 2009; Bettencourt et al., 2007; Nomaler et al., 2014). Empirical evidence shows that larger cities are typically associated with increased rates of human interaction, innovation, economic productivity, and information exchange (Milgram, 1970; Bornstein and Bornstein, 1976; Fujita et al., 2001; Centers for Disease Control and Prevention, 2012; Bettencourt et al., 2008). These findings provide a foundation for analyzing how urban scale shapes various aspects of social and economic life (Pumain, 2004; Kühnert et al., 2006; Bettencourt et al., 2007; Ortman et al., 2015; Lobo et al., 2020). Although urban scaling laws have consistently revealed superlinear growth for various socioeconomic metrics as city size increases, important questions remain about how these patterns emerge and how robust they are under conditions of pronounced socioeconomic inequality and spatially uneven urban development (common in the global south). Such conditions can disrupt access to opportunities, constrain mobility, and reduce the frequency and diversity of social interactions, factors typically assumed to scale uniformly with population in standard models. Moreover, the mechanisms driving these regularities, particularly the role of underlying social networks and how individuals form and maintain connections in larger cities, remain poorly understood (Schlöpfer et al., 2014; Samaniego et al., 2020).

Some studies have used deviations from expected scaling patterns to detect early signs of urban decline, infrastructure stress, or reduced resilience in specific cities, while others have examined how scaling exponents change during periods of rapid urban expansion, suggesting that these laws may also capture transitions between different urban regimes (Garmestani et al., 2008; Levin, 1999; Barnosky et al., 2012). In complex systems such as

cities, interactions among heterogeneous components give rise to flows of information, goods, and energy that underlie emergent scaling patterns (Ribeiro and Rybski, 2023). However, the conventional mean-field perspective, which treats cities as homogeneously interacting populations, risks overlooking the influence of spatial, economic, and social heterogeneities, as well as the existence of loosely connected subsystems within urban areas. Integrating urban scaling with traditional approaches in economics and geography, which emphasize the spatial and behavioral heterogeneity of firms and individuals, will be crucial for elucidating key aspects of underlying urban processes (Glaeser, 2011; Fujita et al., 1999; Bettencourt et al., 2014; Arvidsson et al., 2023). Such a synthesis would also offer new avenues for sustainably managing complex urban adaptive systems (Levin and Clark, 2010). Testing these theoretical frameworks empirically, particularly in diverse socioeconomic contexts, requires large-scale data on actual human interactions.

Recent theoretical work suggests that superlinear scaling patterns may emerge directly from networks of human interactions (Arbesman et al., 2009; Bettencourt, 2013; Pan et al., 2013), often linked to the observation that individuals in larger cities tend to maintain more social contacts on average, suggesting that per capita connectivity increases consistently with urban scale (Bettencourt, 2013). Empirical analyses using mobile phone data indicate that while human mobility is often exploratory, individuals tend to concentrate their daily activities within a limited number of locations (Alessandretti et al., 2018), a pattern reminiscent of cognitive constraints such as Dunbar's number (Ruiter et al., 2011). Despite these advances, most studies of urban interaction networks continue to rely on indirect assumptions, largely because capturing fine-grained data across the full spectrum of interactions is difficult (Blondel et al., 2015a; Yuan et al., 2025). Moreover, it remains unclear whether the mechanisms underlying urban scaling are robust in developing contexts, where inequality, infrastructure disparities (Pandey et al., 2025), and uneven technological adoption may alter the mechanisms that link population size to social connectivity.

The widely accepted notion of increasing returns to scale in larger cities is often linked to higher densities of interpersonal interaction (Ribeiro and Rybski, 2023; Schlöpfer et al., 2014; Samaniego et al., 2020). However, limitations in data availability, particularly in large developing countries, have precluded rigorous testing of this hypothesis. While studies in smaller nations or more data-rich regions have yielded promising results (Schlöpfer et al., 2014; Samaniego et al., 2020), few attempts have validated these scaling relationships in contexts marked by high levels of inequality and significant

regional disparities in infrastructure, economic development, and market access (Xu et al., 2025). In fact, a recent work looking at the scaling of crime (Prieto-Curiel and Menezes, 2025) demonstrates that in Africa, isolation of urban areas appears to be more related to violent crime than population. Hence, mobile phone data offer a unique opportunity to address this empirical gap at scale.

To address this gap, we analyze an exceptionally large Call Detail Records (CDR) dataset: 3.1 billion mobile phone records across Brazilian cities. This vast and heterogeneous dataset enables a rigorous test of how well urban scaling laws hold in a setting of considerable socioeconomic inequality, and allows us to explore whether the observed patterns might generalize to other rapidly growing regions of the world. We develop two complementary methodological approaches to empirically examine the relationship between city size and intensity of human interaction. First, we quantify the average number of intra-company mobile phone contacts per individual as a function of city size, expanding on prior empirical investigations in Portugal, the United Kingdom, and Chile (Schlöpfer et al., 2014; Samaniego et al., 2020). Second, we supplement this perspective by measuring the number of individual trips, positing that patterns of physical mobility capture an additional, often overlooked dimension of urban engagement.

Brazil's cultural, economic, and geographic diversity makes it a compelling setting to evaluate the universality of urban scaling laws. For example, according to the Brazilian Institute of Geography and Statistics (IBGE) in 2020, São Paulo is the largest Brazilian municipality with more than 12 million inhabitants and contributes nearly 32% of the national GDP (approximately US\$ 500 billion). In contrast, Manaus, the seventh-largest municipality with 2.2 million inhabitants, contributes about US\$13 billion of Brazilian GDP. Such disparities reflect the country's extreme levels of regional inequality and raise important questions about whether standard urban scaling relationships remain valid under conditions of uneven infrastructure, technological diffusion, and income distribution.

In this article, we address these empirical gaps by examining scaling properties of social interactions, measured through phone contacts and physical trips, across the 100 most populous Brazilian municipalities. This study relies on official administrative boundaries (municipalities) to delineate urban units, such that city and municipality are used interchangeably to denote each unit. This methodological choice reflects different definitions of metropolitan areas in Brazil, where several large dense urban areas are not recognized as such, making it difficult to manually compile data from different metropolitan areas. Selecting the 100 most populous Brazilian municipalities

allows us to analyze a valuable sample while keeping the number of units tractable. Moreover, for in many small cities in Brazil, municipal boundaries are larger than the urban area, which could introduce additional biases into analysis. We further address key methodological challenges: correcting for market share biases stemming from the uneven presence of a single operator across cities of varying size, developing robust frameworks to estimate aggregate interaction levels from sampled user data, and inferring users' primary residences based on temporal calling patterns. Additionally, we document substantial differences in mobile phone adoption rates, showing that larger cities exceed those in smaller localities. Our analysis reveals that infrastructure, approximated by antenna deployment, scales sublinearly with the actual user base, suggesting the presence of economies of scale in communication network provisioning.

In summary, this study makes six main contributions. First, we provide a national-scale validation of urban scaling laws by analyzing mobile phone records across 100 large Brazilian cities, a data scope rarely attained in previous work. Second, using both phone contacts and physical trips, we demonstrate that human interaction scales superlinearly with city size. Third, we offer methodological refinements for correcting telecom operator market share biases and estimating aggregate interaction levels from sampled data via temporal call-pattern analysis. Fourth, we uncover sublinear scaling of infrastructure when measured against active users rather than total populations, indicating economies of scale in telecommunications. Fifth, we present evidence of higher mobile usage in bigger cities, underscoring how connectivity itself scales with urban size. Lastly, we show that these patterns persist even in the face of profound socioeconomic inequalities, suggesting that universal mechanisms underlie urban growth processes across a range of social and economic contexts. These findings extend our understanding of how scaling laws, infrastructure, and human connectivity intertwine in developing nations, highlighting both the generality and flexibility of scaling principles in shaping urban life.

2 Socio-economics and urban scaling

We start by establishing the theoretical foundation for our analysis by outlining how socioeconomic output scales with city size through human interactions. We first review the general framework connecting socioeconomic metrics to interaction density, then justify our methodological choice to use both mobile phone contacts and physical trips as complementary proxies for measuring urban interactions (Alessandretti et al., 2023).

Recent theoretical work (Ribeiro and Rybski, 2023) connects socioeconomic metrics Y (e.g., GDP) to

interactions between urban dwellers in a city of population size N . This framework proposes that $Y \propto gN(N-1)n_c \sim gN^2n_c$, where n_c is the probability of an encounter occurring between any pair of urban dwellers, g denotes the socioeconomic output generated from the encounter between such pairs, and $N(N-1)$ represents the total number of possible pairs. The central distinction among competing models of socioeconomic scaling lies in how n_c is estimated. For instance, Bettencourt (2013) models $n_c \equiv a_i/A_n$, where a_i is the individual's area of movement, and A_n is the total infrastructural urban area. Alternative approaches define n_c as the contact density, expressed as $n_c = \langle k_i \rangle / N$, where $\langle k_i \rangle$ represents the average number of contacts, *the degree*, of the individuals in the city.

In this study, we adopt this density-based approach. Assuming that g is scale-invariant, the socioeconomic output can be expressed as:

$$Y \propto N \langle k_i \rangle. \quad (1)$$

This formulation highlights that if the total number of contacts in the city ($N \langle k_i \rangle$) scales superlinearly with population, then the superlinear scaling of socioeconomic urban metrics emerges as a direct consequence. Cities facilitating greater social interactions among urban dwellers tend to generate higher socioeconomic output, here represented by Y .

To operationalize this theoretical framework, we employ two complementary empirical measures of interaction density. First, we estimate contact density using Call Detail Records (CDRs) derived from mobile phone data, which capture remote communication patterns. Second, we measure physical mobility through trips inferred from the same dataset. While phone contacts reflect communication-based interactions, trips provide insights into face-to-face encounters and location-based activities that may have distinct economic implications. This dual measurement strategy allows us to assess whether different interaction modalities exhibit similar scaling behaviors, thereby testing the robustness of the theoretical predictions.

The use of trips as a measure of interaction density warrants particular justification, as physical travel represents a more costly and deliberate form of interaction compared to remote communication. A more precise definition of what constitutes a “trip” will be presented in the following sections. The primary motivation for including trips alongside phone contacts lies in three factors:

1. **Cost-benefit considerations:** Physical travel involves substantial costs in time, money, and energy, implying that such interactions are typically of higher perceived value (Aguilera, 2008).

2. **Interaction quality:** Historically, face-to-face meetings, enabled by physical travel, often facilitate more substantial economic activities than remote communications (Storper and Venables, 2004; Bathelt et al., 2004).
3. **Correlation with economic activity:** Trip patterns frequently correspond to business activities, professional networking, and consumer behavior, providing direct insights into wealth generation mechanisms (Duranton and Turner, 2012).

This methodology captures high-value interaction densities, particularly consequential for economic output and agglomeration effects, complementing traditional analytical approaches.

We now turn to the data and methods used to implement this framework empirically. Section 3 describes the Call Detail Records dataset, the inference procedures for identifying user residences, and the operational definitions of contacts and trips.

3 Data and methods

3.1 Call Detail Records (CDR) data and Brazilian municipalities

Call Detail Records (CDRs) are widely used to study human mobility and social interactions due to their high spatial and temporal resolution and broad population coverage (Blondel et al., 2015b; Gonzalez et al., 2008; Barbosa et al., 2018). Each record typically includes meta-data about the time, duration, origin and destination of a call, and the location of the antenna that routed it. These data have been instrumental in inferring mobility patterns, social connectivity, and urban behaviour in diverse contexts.

We analyze a CDR dataset containing 3.1 billion records from Brazil's 100 most populous cities, covering the period between March 13 and April 13, 2013. The data were provided by a single national telecom operator and correspond to a typical, non-eventful month, minimizing the influence of holidays or special events on interaction patterns. For contractual reasons, the specific operator cannot be identified, and city-level market share data cannot be disclosed. However, the operator maintains an average market share of approximately 20% across Brazil's cities, with variation around this average. Importantly, there is no systematic relationship between city size and the operator's penetration rate, meaning that sampling patterns do not vary systematically with population size. At the time of data collection, voice calls were the primary communication mode, as Voice-over-IP (VoIP) had not yet reached widespread adoption. In this context, 85% of urban Brazilians utilized mobile phones. The dataset only includes outgoing calls from users on

the operator's network and excludes SMS or internet traffic. All data were anonymized prior to analysis. (See Appendix A for a description of raw CDR dataset).

Each call is associated with a specific antenna, from which a Voronoi tessellation is constructed to approximate its coverage area (see Fig. 7). A toy-example image is presented in Fig. 1. Owing to varying antenna densities, these Voronoi polygons can differ significantly in size, leading to decreased positional precision in sparsely covered regions. Consequently, while these partitions provide a practical basis for spatial analysis, potential bias may exist in areas with limited antenna coverage.

Brazil has 5,570 municipalities, ranging from megacities with over 10 million residents, supported by numerous antennas, to small towns of only a few thousand

inhabitants served by minimal telecom infrastructure. This heterogeneity makes Brazil an ideal setting for investigating mobility and social interactions at a national scale. However, analysing all 5,570 municipalities can dilute the reliability of scaling estimates, particularly for smaller localities lacking robust data. We therefore restrict our study to the 100 most populous municipalities, whose collective diversity spans multiple climates, cultures, and economic conditions.

Population size for each of these 100 municipalities was obtained from the 2010 census conducted by the Brazilian Institute of Geography and Statistics (IBGE), the nation's official provider of demographic and socio-economic statistics (IBGE, 2025). Given the role of population as a core variable in urban scaling analysis, we

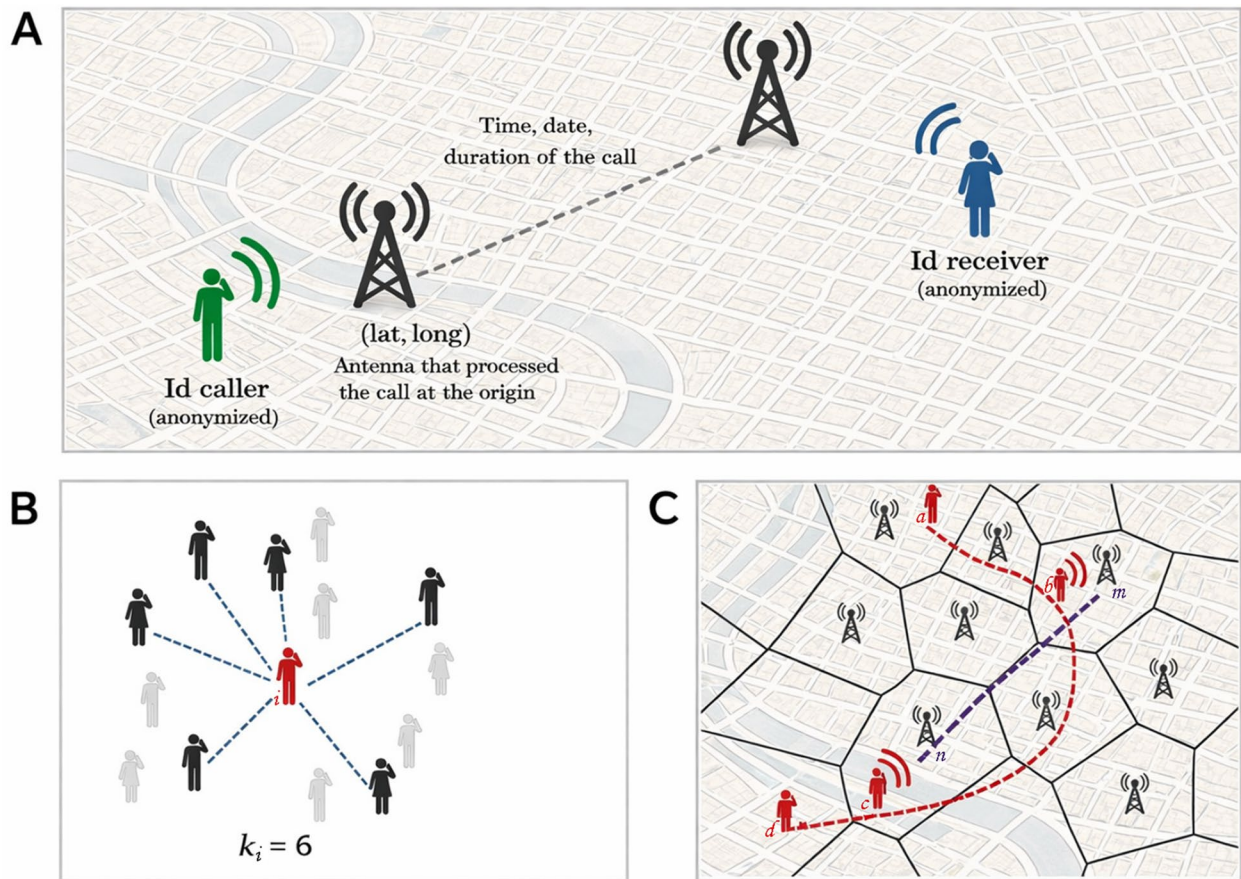


Fig. 1 CDR structure and key definitions. **A** A single call made by a caller to a receiver registered in the dataset is displayed. All available information related to this call is shown, which includes the location of the antenna that processed the call at the origin, call duration, timestamp, and the anonymized IDs of both the caller and the receiver (more details in Appendix A). **B** From the dataset, it is possible to infer the degree k_i of the i -th user, defined as the number of distinct users with whom user i has made at least one call. In this representative case, $k_i = 6$. **C** Schematic representation of a "trip". The red dashed line represents the actual trajectory of a user moving from point a to point d , which is not directly accessible in the dataset. During this movement, this user performs a call in b and then another call in c . This movement is considered a trip if the time interval between these two calls has a minimum of 30 minutes and a maximum of 4 hours, and if the distance $\overline{mn}$ between the antennas (purple dashed line) where the calls are received is larger than 2km. This panel also presents the Voronoi cells created from the spatial tessellation of the antennas

supplemented the CDR dataset with IBGE-based figures for consistency and comparability. The complete raw dataset listing of cities, population size (N), and users (U) is available in an external repository: <https://doi.org/10.5281/zenodo.15359949>. See Fig. 4 for their geographic locations.

By limiting our analysis to Brazil's 100 largest cities, we achieve a balance between examining a sufficiently large and diverse sample of urban centers while retaining robust statistical power in each locale. This enables us to examine whether conventional insights into urban scaling and mobility are applicable in one of the world's most heterogeneous and unequal societies. The combination of detailed CDR data and comprehensive census information provides a rich, representative snapshot of everyday mobility and communication, laying the groundwork for subsequent analyses of how city size, infrastructure, and social connectivity interrelate.

3.2 Inference of people's residences

Understanding individual mobility patterns requires a baseline from which to evaluate daily travel and social interactions, making the identification of users' residences critical. By establishing where individuals return to or spend their nights, we can anchor movement data to a specific "home" location. This helps differentiate residents from transient visitors and clarifies commuting behaviors and other regular travel patterns that underlie broader socioeconomic processes.

We estimate the presumed residence of each user as the antenna location with the highest frequency of calls placed between 7 p.m. and 6 a.m. on weekdays and at any time on weekends. To enhance the accuracy of this estimation, at least 50% of all calls made during these periods must originate from the identified antenna. Only users with a clearly established residential location based on these criteria were included in the analyses. We note that this residence inference procedure may be affected by differential calling patterns across cities. If users in larger cities place calls more frequently, they may be more likely to meet the 50% threshold criterion, potentially contributing to the observed superlinear relationship between the number of identified users (U) and population (N). We address this in Sect. 4. This approach aligns with prior research that successfully employs mobile phone data to infer home locations (Ahas et al., 2010; Isaacman et al., 2011; Vanhoof et al., 2018).

3.3 Definition of contacts and trips

To capture both remote and face-to-face dimensions of social interaction, we adopt two complementary measures derived from the CDR dataset: the number of voice-call *contacts* and the number of *trips*.

We define the number of contacts by constructing a network whose nodes represent individual users, while the presence of an edge signifies at least one voice call between two users. Specifically, user j is considered a *contact* of user i if user i has placed one or more calls to j during the study period. For each node (i.e., user), the total number of distinct contacts can be interpreted as the individual's effective communication reach via voice calls. This approach follows established network-based analyses of mobile phone datasets, where edges denote social ties inferred from call activity (Blondel et al., 2015b; Onnela et al., 2007) (see Fig. 1 for a visual representation).

While voice contacts capture remote communication, they may not fully reflect the face-to-face and economic activities typically requiring physical presence. To account for these, we define a *trip* as the movement between two locations exceeding a minimum distance threshold (between antennas) within a given time window. We set an antenna distance threshold of 2 km occurring within at least 30 minutes and up to 4 hours, thereby reducing the risk of overcounting minor displacements or signal fluctuations in densely populated areas with numerous antennas. These specific parameters (2 km distance, 30 minutes–4 hours time window) were previously validated for this operator's data in the Rio de Janeiro Metropolitan Area (Barboza et al., 2021), where trip estimates derived using these thresholds aligned closely with contemporaneous household travel surveys. This criterion focuses mainly on medium- to long-range trips commonly associated with work-related journeys, professional engagements, and significant consumer activities, while inevitably under-representing short pedestrian trips. Figure 1-C provides an illustrative example of how a trip is defined.

To implement this definition, we develop origin–destination (OD) matrices, following a well-known procedure in the literature (Alexander et al., 2015; Barboza et al., 2021; Toole et al., 2015). In particular, we build on the work of Calabrese et al. (2013), which infers user movement by examining two consecutive phone calls made from distinct locations (i.e., different antennas). By identifying the origin and destination of each such inferred journey, we form an OD matrix whose entries represent aggregate travel flows between spatial units. More details on the trip detection algorithm, including handling of potential boundary cases and validation steps, are provided in Appendix E.

Together, these two metrics serve as proxies for the multifaceted ways in which urban residents interact. Contacts capture communication-based ties, while trips provide insights into physical mobility and face-to-face engagements. Examining both measures in parallel

allows us to distinguish patterns that may be driven by remote connectivity from those underpinned by in-person encounters.

3.4 Estimation of city-level interaction metrics from sampled data

The Call Detail Records dataset provides information of a single telecommunications operator, representing a sample of the total urban population. To estimate city-level interaction metrics from this sampled data, we require a mathematical framework that extrapolates from observed users to the entire population. Here we establish the estimation procedures for two key quantities: the total number of contacts (cumulative degree) and the total number of trips in each city.

3.4.1 Estimating the cumulative degree

Our goal is to estimate the level of interaction in cities of various sizes, specifically through the average number of mobile phone contacts per person ($\langle k_i \rangle$) and the total number of contacts (K) in a specific city. To formalize these metrics, let k_i denote the number of contacts (i.e., degree) of an individual i . Then, the average degree in a city of population N is

$$\langle k_i \rangle = \frac{1}{N} \sum_{i=1}^N k_i, \quad (2)$$

and multiplying by the total population yields the city-wide cumulative degree:

$$K = N \langle k_i \rangle. \quad (3)$$

Computing these quantities directly would require knowing k_i for every single inhabitant, which is unfeasible. However, CDRs allow us to estimate k_i by tracking the number of distinct mobile phone contacts associated with each user in our sample. Specifically, we define

$$\langle k_i^s \rangle = \frac{1}{U} \sum_{i=1}^U k_i^s, \quad (4)$$

where k_i^s is the number of voice-call contacts of the sampled user i with other sampled users residing in the same city, and U is the total number of sampled users. Consequently,

$$K^s = U \langle k_i^s \rangle \quad (5)$$

gives the cumulative degree restricted to the subset of users for whom CDR data are available. Note that $\langle k_i^s \rangle$ and K^s can be directly computed from the data, but $\langle k_i \rangle$ and K for the full population of size N must be inferred.

To reconcile these metrics, we adopt the assumption

$$\langle k_i^s \rangle \approx \langle k_i \rangle, \quad (6)$$

following arguments also made by Schlöpfer et al. (2014), under the rationale that an individual's propensity to maintain contacts does not depend strongly on the share of users captured in the sample. As demonstrated in Appendix D, the average degree for a given city converges as the sample size becomes sufficiently large. This convergence supports the assumption that the average degree computed from the set of users (which is empirically accessible) reliably approximates the average degree of the entire population (which is not directly observable), thereby justifying the assumption stated in Eq. 6.

Then, using the assumption in Eq. 6 along with Eqs. 3 and 5 to express the cumulative degree of the entire city in terms of the cumulative degree of the sample:

$$K \sim \left(\frac{N}{U} \right) K^s, \quad (7)$$

which, in turn, leads us to

$$K \sim N^{1+\beta_0\beta_U}, \quad (8)$$

given the exponents defined in Table 1. Plugging in the empirically obtained $\beta_0 = 1.29$ and $\beta_U = 0.15$ yields $K \sim N^{1.19}$, signaling a clear superlinear regime.

3.4.2 Estimating the number of trips

To complement our analysis of phone-based contacts, we also estimate the total number of trips within each city. We define an average number of trips per person, $\langle T_i \rangle$, as follows:

$$\langle T_i \rangle = \frac{1}{N} \sum_{i=1}^N T_i, \quad (9)$$

where T_i is the number of trips undertaken by the i -th city resident over the specified observation period. From this, we obtain the total number of trips τ in the city by multiplying $\langle T_i \rangle$ by the population N :

$$\tau = N \langle T_i \rangle, \quad (10)$$

analogous to the cumulative degree.

In practice, $\langle T_i \rangle$ and τ are not directly observable from raw CDR data. We therefore rely on a sampling strategy analogous to the approach used for contact estimation. Specifically, let T_i^s denote the number of trips made by user i in our sample of size U . We can then directly compute:

$$\langle T_i^s \rangle = \frac{1}{U} \sum_{i=1}^U T_i^s, \quad (11)$$

which in turn yields the total trips in the sample:

Table 1 Scaling exponents for urban variables from CDR analysis

Variable	Explanation	Scaling relation	Value of β	Std. error	Observation
GDP	Gross Domestic Product	$GDP \sim N^{\beta_G}$	$\beta_G = 1.12$	0.08	—
U	Number of Users	$U \sim N^{\beta_0}$	$\beta_0 = 1.29$	0.03	—
U/N	Ratio of Users to Population	$U/N \sim N^{\beta_0-1}$	$\beta_0 - 1 = 0.29$	0.03	—
N_A	Number of Antennas	$N_A \sim N^{\beta_A}$ $N_A \sim U^{\beta_A/\beta_0}$	$\beta_A = 1.11$ $\beta_A/\beta_0 = 0.86$	0.06 0.05	Empirical estimation: $N_A \sim U^{0.84 \pm 0.04}$ (see Fig. 2c)
$\langle k_i^s \rangle$	Avg. Degree in the Sample	$\langle k_i^s \rangle \sim U^{\beta_U}$	$\beta_U = 0.15$	0.01	—
K^s	Cumulative Degree in the Sample	$K^s \sim U \langle k_i^s \rangle$ $K^s \sim U^{1+\beta_U}$	$1 + \beta_U = 1.15$	0.01	—
K	Cumulative Degree in the City	$K \sim N^{\beta_c}$	$(1 + \beta_0 \beta_U) \rightarrow \beta_c$ (Eq. 8) $\beta_c = 1.19$	0.02	β_c is not Accessible empirically
$\langle T_i^s \rangle$	Avg. # of Trips in the Sample	$\langle T_i^s \rangle \sim U^{\beta_T^s}$	$\beta_T^s = 0.19$	0.04	—
τ^s	Total # of Trips in the Sample	$\tau^s \sim U \langle T_i^s \rangle$ $\tau^s \sim U^{1+\beta_T^s}$	$1 + \beta_T^s = 1.19$	0.04	—
τ	Total # of Trips in the City	$\tau \sim N^{\beta_\tau}$	$(1 + \beta_0 \beta_T^s) \rightarrow \beta_\tau$ (Eq. 10) $\beta_\tau = 1.25$	0.05	β_τ is not Accessible empirically

Estimated scaling exponents for social interactions and trips, derived from Call Detail Records (CDR) covering Brazil's 100 most populous cities. Social interactions are measured by the number of contacts within the voice-call network, and the number of trips are inferred from users' call sequences. Slopes (β) are obtained via ordinary least squares (OLS) regression of the logarithmic transformation (log-log plot) of the variables under analysis. Here we use # as a simplified notation for number

$$\tau^s = U \langle T_i^s \rangle. \quad (12)$$

As before, τ for the entire city is unavailable, so we estimate it by assuming $\langle T_i \rangle \approx \langle T_i^s \rangle$. This assumption follows the rationale that each individual's trip behavior should not depend on the fraction of the population captured in the sample. Combining this assumption with Eqs. (10) and (12), we get

$$\tau \sim \left(\frac{N}{U} \right) \tau^s, \quad (13)$$

and by incorporating the scaling relations described in Table 1, we obtain

$$\tau \sim N^{1+\beta_0 \beta_T^s}. \quad (14)$$

Substituting the empirically determined exponents $\beta_0 = 1.29 \pm 0.03$ and $\beta_T^s = 0.19 \pm 0.04$, we arrive at $\tau \sim N^{1.25}$ ($\beta_\tau = 1.25 \pm 0.05$). This result indicates a clear superlinear relationship between the total number of trips and population size, mirroring the pattern observed for phone-based contacts. The empirical results of applying this estimation framework are presented in the following section.

4 Results

Our empirical analysis addresses three primary research questions derived from the theoretical framework presented in Sect. 2: (1) How does the infrastructure

required for telecommunications scale with city size? (2) Do social interactions, measured through both remote communication and physical mobility, scale superlinearly with population? (3) Are these scaling patterns robust across cities with substantial socioeconomic heterogeneity? We address these questions through systematic scaling analysis of multiple urban variables derived from our national-scale CDR dataset, employing ordinary least squares regression on log-transformed data to estimate scaling exponents and their associated uncertainties.

Results are summarized in Table 1. All urban variables analyzed, including GDP, the number of antennas (N_A), and the number of mobile phone users (U) of the telecommunications provider that supplied the data, are presented alongside their respective scaling equations and empirical exponents (β). These exponents quantify how each variable changes with city size, as represented by i) the census population or ii) the number of users. The slopes are estimated via unweighted ordinary least squares (OLS) regression applied to the logarithmic transformations (log-log plot) of the variables under analysis. The table also includes standard errors and relevant notes about each variable.

Table 1 shows a superlinear relationship of GDP with population ($\beta_G = 1.12$). This aligns with previous observations showing that the sample of 100 most populous municipalities behaves as expected for larger cities, yielding disproportionately higher economic

output (Lobo et al., 2013). Additionally, the number of users (U) grows more steeply ($\beta_0 = 1.29$) than the overall population, indicating that phone adoption intensifies as cities expand. The number of antennas (N_A) exhibits a superlinear relationship with population size ($\beta_A = 1.11$), which contradicts previous findings Bettencourt et al. (2007) that suggest infrastructure-related variables like this one should scale sublinearly. However, the number of antennas exhibits a sublinear relationship when expressed relative to the number of users, with a scaling exponent of 0.84. We further explore this apparent contradiction in the following subsections.

The latter part of the table focuses on social interactions (both *contacts* and *trips*) as inferred from call detail records (CDR). The cumulative degree K of the voice-call network shows superlinear scaling with population ($\beta_c = 1.19$), implying that larger cities are associated with more contacts per person. Similarly, the total number of trips in the city, τ , also scales superlinearly with population ($\beta_\tau \approx 1.25$), reflecting a heightened level of physical mobility in bigger urban centers. Each of these findings will be discussed in further detail in the following subsections, where we explore how they contribute to our broader understanding of urban scaling and social interaction patterns.

4.1 Number of phone ID, population size, and adoption ratios

Our analysis reveals a superlinear relationship between U (the number of mobile phone IDs) and N (the total population) (Fig. 2). Throughout this analysis, the term “users” and the variable U refer to unique mobile phone IDs (SIM cards) in the operator’s network rather than

individual persons, recognizing that individuals may possess multiple IDs. This finding challenges the assumption of a near-linear relationship between population size and mobile phone usage at a large scale. Instead, it indicates that mobile phone adoption increases disproportionately faster than population growth in larger urban areas. Specifically, larger cities not only have more residents but also have increasingly common use of more than one ID per capita. By 2013, most mobile devices supported multiple phone IDs, and in Brazil, the prevalence of pre-paid plans may have led some users to maintain multiple phone IDs across different operators, though our single-operator data cannot directly confirm this hypothesis. One plausible explanation for this phenomenon is that denser, more economically dynamic urban environments foster social and business conditions conducive to the adoption of multiple phone IDs and peer-driven usage patterns. Additionally, these findings suggest the presence of feedback loops in large cities, where the perceived necessity for mobile connectivity accelerates adoption rates. Such feedback mechanisms may influence broader urban dynamics, including the demand for communication infrastructure and the intensity of social interactions, as expanding user bases amplify the value of real-time digital connectivity.

4.2 Number of antennas

While the number of antennas across cities is typically viewed as an infrastructure variable that usually scales sublinearly with population size (Bettencourt et al., 2007; Kühnert et al., 2006; Meirelles et al., 2018; Ribeiro and Netto, 2025), our work shows that the number of antennas (N_A) is superlinear ($\beta_A = 1.11 \pm 0.06$), implying that larger populations are served by a disproportionately

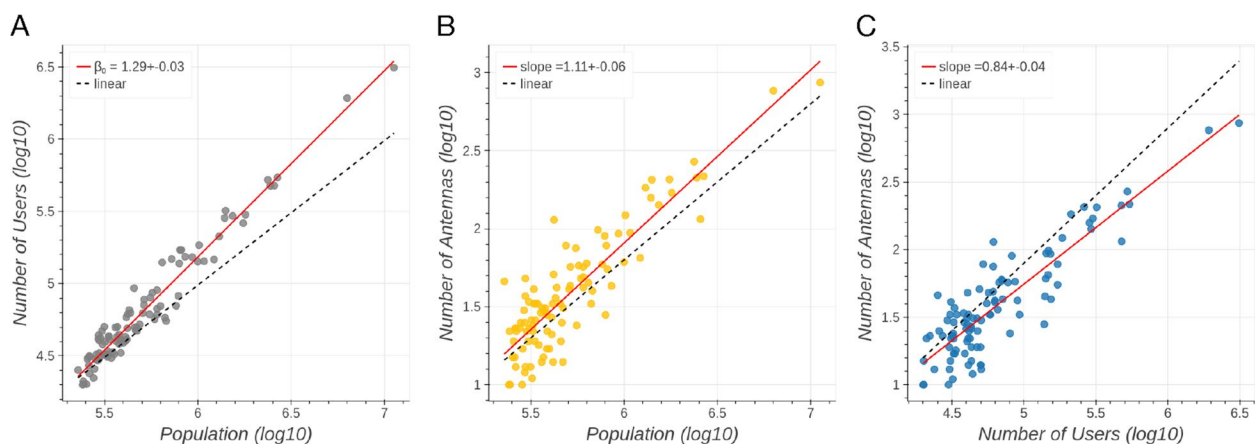


Fig. 2 Relations between number of Users (U), Population (P), and number of Antennas (N_A). **A** U exhibits superlinear behavior in relation to N . This non-linearity between U and N results in distinct scaling regimes for the number of antennas (N_A): **B** superlinear when the number of antennas is considered relative to population size; and **C** sublinear when considered relative to the number of users. Dashed lines indicate the one-to-one linear relationship, where the slope is equal to 1.0

larger number of antennas. In contrast, when using the number of mobile phone users (U) as a proxy for city scale, the number of antennas becomes sublinear ($N_A \sim U^{0.84}$), consistent with prior findings for infrastructural scaling. This subsection outlines potential explanations for the observed patterns.

At least two explanations may account for the observed superlinear scaling of the number of antennas (N_A) with respect to population size (N). The first is that larger cities, characterized by more diverse and complex activity patterns, may require operators to deploy additional antenna sites to ensure adequate service quality and coverage. The second explanation is that this trend could reflect the tendency of operators to prioritize or aggressively expand infrastructure in larger cities, where the potential for higher returns on investment and user demand is greater.

Regarding the sublinear relationship between the number of antennas (N_A) and the number of users (U), one possible explanation is that denser clusters of users enable more efficient utilization of shared antenna capacity, reducing the need for proportional infrastructure expansion and improving cost-effectiveness. Additionally, operators may strategically prioritize antenna deployment based on actual device usage patterns rather than relying solely on city population size. This observation supports the hypothesis that companies adjust their infrastructure investments in response to user demand dynamics rather than population metrics alone. Furthermore, the sublinear scaling of N_A with respect to U may reflect economies of scale in network deployment, an effect that is not evident when considering population size (N) as the primary variable. This suggests that user density and demand play a critical role in shaping infrastructure investment strategies, potentially leading to more efficient resource allocation in urban environments.

4.3 Nominal/effective population

Collectively, these findings highlight how the choice of city-size metric can significantly influence our understanding of infrastructure scaling. This apparent discrepancy in the scaling of the number of antennas, superlinear with respect to population but sublinear with respect to the number of users, reveals a crucial nuance in interpreting urban scaling: the nominal population may not always be the most appropriate variable for the phenomenon being studied. Since the number of users scales nonlinearly with population size, relying solely on population-based metrics can produce misleading scaling exponents. Incorporating user-based measures is therefore essential to accurately capture infrastructure dynamics and avoid erroneous conclusions.

In our context, the number of phone IDs (U) serves as an effective population for telecommunications, reflecting not the number of individuals but the intensity and multiplicity of their connectivity. When we reframe the analysis using U as the effective population, we find that antenna deployment scales sublinearly with this user base. This aligns with the efficiency expectations of urban scaling theory (Bettencourt et al., 2007) and confirms that infrastructure benefits from economies of scale, even as the effective user population grows superlinearly with the nominal population. These insights emphasize the importance of carefully defining population variables in scaling analyses. For many urban processes, especially those involving technology adoption and connectivity, the raw count of residents may be less informative than measures of effective participation.

However, the superlinear relationship between phone IDs and population warrants cautious interpretation and further investigation with multi-operator datasets to determine whether this pattern would still be present when multiple operators are considered. We continue to be confident on the results because the operator's market share shows no systematic variation with city size, suggesting that market penetration does not drive this relationship.

4.4 Scaling of social interactions

Having established the scaling patterns of phone adoption and infrastructure, we now present our main findings on how social interactions scale with city size. As described in Sect. 3, we apply the estimation framework to quantify both mobile phone contacts and physical trips across the 100 cities.

Our analysis reveals that the cumulative degree K scales superlinearly with population ($\beta_c = 1.19$), as shown in Fig. 3. To illustrate, in São Paulo ($\sim 10^7$ inhabitants), the average user maintains roughly 10 mobile phone contacts, whereas in Belém ($\sim 1.3 \times 10^6$ inhabitants), this figure drops to about 7 contacts. This pattern aligns with earlier studies in Portugal and Chile (Samaniego et al., 2020; Schlöpfer et al., 2014), which also find superlinear growth in the number of social contacts. However, our results expand the literature by applying these methods to a national-scale dataset spanning highly diverse socioeconomic and geographic conditions, reinforcing the universality of superlinear interaction growth across larger urban populations.

Similarly, our analysis shows that the total number of trips τ scales superlinearly with population ($\beta_\tau = 1.25$), as also presented in Fig. 3. This result mirrors the pattern observed for phone-based contacts, indicating that larger cities have a disproportionately higher volume of travel

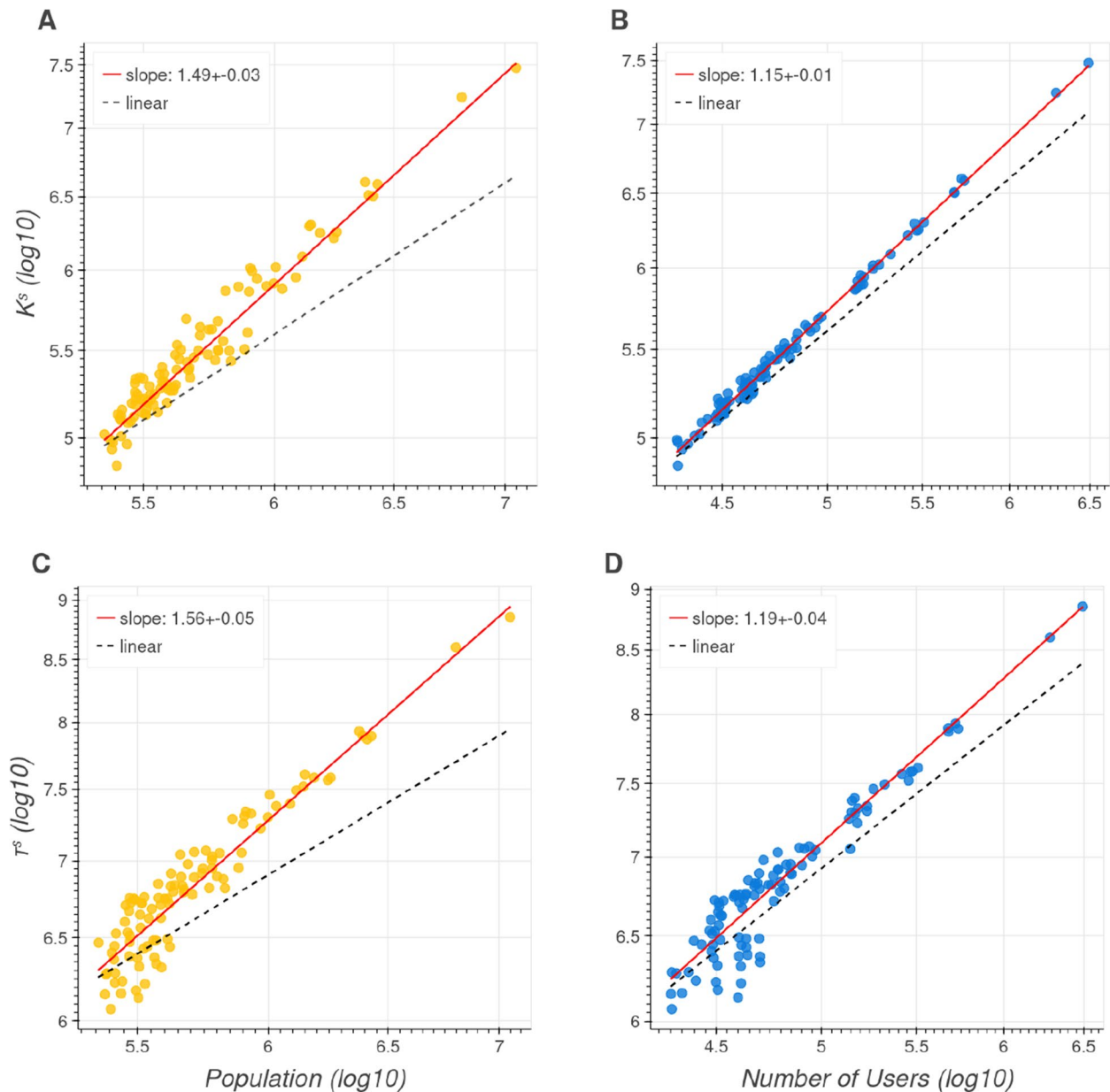


Fig. 3 Scaling of degree and trips. (Top): Cumulative degree in the dataset (K^s) as a function of **A** city population size (N) and **B** the number of users (U). (Bottom): Number of trips restricted to the dataset (τ^s) as a function of **C** population size and **D** the number of users. Note that the superscript 's' represents the "sample" estimation. Dashed lines indicate the one-to-one linear relationship, where the slope is equal to 1.0

and suggesting that increased physical mobility may be reinforcing the same agglomeration dynamics observed in communication networks. Further discussion of the implications and contrasts between trips and contacts can be found in Sect. 5.

4.5 Robustness of scaling estimates

The aforementioned scaling relationships reported are based on OLS regression of log-transformed variables

across 100 cities. To assess the robustness of our findings, we conducted several validation analyses. First, as detailed in Appendix D, we verified that the average degree ($\langle k_i^s \rangle$) converges as sample size increases, supporting our assumption that sampled quantities reliably approximate population-level metrics (Eq. 6). This convergence holds even for smaller cities in our sample, such as Palmas, demonstrating that our methodology is not biased toward larger urban centers.

Second, we note that all reported scaling exponents show relatively small standard errors (Table 1), indicating stable estimates across the 100-city sample. The consistency between independently measured quantities further supports our findings: both contacts ($\beta_c = 1.19$) and trips ($\beta_\tau = 1.25$) exhibit similar superlinear scaling exponents, suggesting that the observed patterns reflect genuine urban processes rather than measurement artifacts.

Third, our finding that antenna deployment scales sub-linearly with the number of users aligns with established infrastructure scaling theory (Bettencourt et al., 2007), providing external validation that our analytical framework captures known urban phenomena. The superlinear scaling of GDP ($\beta_G = 1.12$) likewise corroborates previous studies in other national contexts (Lobo et al., 2013), suggesting that our results are consistent with broader empirical patterns in urban science.

While our analysis relies on data from a single telecommunications operator, the consistency of our findings with prior studies using different operators in different countries (Sect. 5) provides additional validation that the observed scaling patterns are not artifacts of operator-specific market dynamics.

Finally, the geographic, economic, and climatic diversity of the 100 Brazilian cities in our sample, spanning from Manaus in the Amazon to Porto Alegre in the temperate south and from wealthy São Paulo to less affluent municipalities, provides a robust test of scaling relationships under heterogeneous conditions. The persistence of clear scaling patterns across this diversity strengthens confidence that these relationships reflect fundamental urban mechanisms rather than context-specific factors.

5 Discussion

In this work, we analyze the scaling laws of human interaction by comparing urban indicators to social variables derived from Call Detail Records (CDRs) from mobile phone interactions across Brazil's 100 most populous cities.

Our findings demonstrate that as city size increases, the per capita number of contacts scales as $\langle k_i \rangle \sim N^{0.19}$ and the per capita number of trips scales as $\langle T_i \rangle \sim N^{0.25}$. This means that residents of a city with 10 million inhabitants maintain approximately 50–55% more mobile phone contacts and 70–80% more trips per capita than residents of a city with 1 million inhabitants, indicating that individuals in larger urban centers foster substantially more extensive interaction networks than those in smaller ones.

This observation strongly aligns with scaling theory (Bettencourt, 2013; Ribeiro and Rybski, 2023), which posits that increased population density enhances the probability of encounters, thereby generating agglomeration effects (e.g., Duranton & Puga, 2004, 2014) and

amplifying opportunities for social, economic, and cultural exchanges. Our large-scale CDR-based analysis provides empirical validation for these theoretical underpinnings across a broad range of urban scales, offering a granular perspective on how individual-level interaction patterns contribute to macroscopic urban dynamics.

To quantify human interaction, we employ two proxies: the average number of mobile phone contacts, $\langle k_i \rangle$, and the number of trips, $\langle T_i \rangle$. While mobile phone contacts reflect remote communication, trips capture activities dependent on physical movement. Our analysis reveals that both metrics exhibit similar superlinear scaling trends, supporting the hypothesis that larger cities foster more intense interaction patterns. Although each proxy has limitations (i.e., mobile phone contacts may overlook face-to-face interactions, and trips may not fully capture brief or virtual exchanges), their combined and consistent scaling behavior provides a comprehensive understanding of urban connectivity, underscoring the pervasive nature of intensified interaction in larger urban environments.

Our results provide compelling evidence that agglomeration effects manifest across both digital and physical interaction layers, strongly reinforcing urban scaling theory. The observed superlinear scaling of both mobile phone contacts ($\beta_c = 1.19$) and physical trips ($\beta_\tau = 1.25$) with city size demonstrates that larger cities facilitate intensified interactions through multiple channels simultaneously. This dual manifestation suggests that digital connectivity and physical mobility may be complementary rather than substitutable mechanisms for urban interaction, consistent with theoretical predictions that cities function as interaction multipliers (Bettencourt, 2013; Pan et al., 2013).

The scaling exponents for contacts ($\beta_c = 1.19$) and trips ($\beta_\tau = 1.25$) are statistically indistinguishable given their standard errors (0.02 and 0.05 respectively), suggesting that the fundamental drivers of agglomeration, that is, the increased probability of productive encounters in denser environments, operate consistently across interaction modalities. However, the slightly higher exponent for trips may indicate that physical mobility is even more responsive to city size than remote communication, possibly because trip-making requires overcoming higher barriers (time, cost, effort) and thus intensifies more dramatically when urban density creates stronger incentives for face-to-face interaction. The superlinear scaling of GDP ($\beta_G = 1.12 \pm 0.08$) lies intermediate between linear scaling and the scaling of interactions ($\beta_c = 1.19$, $\beta_\tau = 1.25$), consistent with the hypothesis that enhanced connectivity creates conditions for economic productivity rather than being a simple byproduct of wealth.

Our work extends previous findings from smaller-scale studies (Samaniego et al., 2020; Schlöpfer et al., 2014) in three key ways. First, we conduct a national-scale analysis of Brazil's largest cities, a geographical scope rarely attained in previous work, uniquely testing urban scaling laws in a context marked by substantial socioeconomic inequality and infrastructural diversity. Second, we incorporate physical mobility (trips) alongside communication patterns as dual measures of interaction, demonstrating that both modalities exhibit similar superlinear scaling. Third, we show that telecommunication infrastructure scales sublinearly with active users ($N_A \sim U^{0.84}$) despite appearing superlinear with nominal population, introducing the concept of effective population for infrastructure analysis. This infrastructure efficiency may partially explain how larger cities support intensified interaction patterns: the telecommunications network becomes more efficient at scale, potentially reducing costs per interaction and thereby facilitating superlinear growth in contacts.

The robustness of scaling relationships across Brazil's diverse urban landscape is itself a substantive finding. Despite enormous variation, where São Paulo contributes approximately US\$500 billion to national GDP with 12 million inhabitants whereas Manaus contributes only US\$13 billion with 2.2 million inhabitants (a per capita GDP difference of factor 2), the scaling relationships remain remarkably consistent. The relatively small standard errors reported in Table 1 and visual inspection of Figs. 2 and 3 confirm that cities across the wealth and size spectrum adhere to the same scaling relationships. This consistency across profound inequality suggests that the mechanisms generating superlinear interaction scaling operate independently of local economic conditions, potentially representing fundamental constraints or opportunities inherent to urban agglomeration itself.

However, several limitations warrant discussion. The cross-sectional nature of our data limits our ability to determine causal relationships between interactions and economic growth. While our findings demonstrate that both social contacts and mobility patterns scale superlinearly with population size, offering a potential explanation for enhanced socioeconomic performance in larger cities, it remains unclear whether heightened social connectivity is the cause or consequence of urban wealth. Future longitudinal studies tracking individual cities over time would complement our analysis by examining how scaling relationships evolve as cities grow (Bettencourt et al., 2020; Ribeiro et al., 2020) and could help disentangle these causal mechanisms.

Our dataset exhibits potential biases that may affect scaling exponent precision. Call patterns show significant temporal variation with pronounced peaks during working hours, and the dataset predominantly consists of prepaid SIM cards (roughly 85%), which are often associated with lower-income demographics and may reflect lower call frequencies. Short pedestrian trips are likely under-represented due to distance and time thresholds required for trip detection. While we address these limitations by aggregating features at the city level, residual biases may persist.

A critical limitation is our reliance on data from a single mobile phone operator, which may introduce sampling biases from city-specific marketing campaigns, pricing strategies, or targeted infrastructure investments. However, several lines of evidence suggest that these operator-specific factors do not substantially distort our main findings. As mentioned in Sect. 3, the operator's market share (approximately 20% on average) shows no systematic relationship with city size, ensuring that sampling patterns do not vary systematically with the independent variable. Also, previous work using this operator's data (Barboza et al., 2021) validated the trip detection parameters against household travel surveys, providing empirical support that aggregate mobility patterns from single-operator data can be representative.

Our findings also align closely with prior studies using different operators in different countries. The superlinear scaling of contacts ($\beta_c = 1.19$) matches results from Portugal and Chile (Samaniego et al., 2020; Schlöpfer et al., 2014) despite different operators and methodologies. The sublinear infrastructure scaling with users ($N_A \sim U^{0.84}$) aligns with established theory (Bettencourt et al., 2007), and GDP scaling ($\beta_G = 1.12$) corroborates prior work (Lobo et al., 2013). This consistency across multiple independent validations suggests that observed patterns reflect genuine urban processes rather than operator-specific artifacts, in spite of covariates that emerge from differences in cities (e.g., socioeconomics, infrastructure, climate).

Nevertheless, we acknowledge that certain biases could affect precise scaling exponent values. If the operator disproportionately attracts high-contact individuals in larger cities due to network effects, or if the prepaid/postpaid user ratio (prepaid users constitute 85% of our sample) varies systematically with city size and these groups exhibit different behaviors, scaling exponents could be affected. We mitigated short-term fluctuations by analyzing a typical, non-eventful month, but future studies integrating data from multiple

operators over extended timeframes remain essential for refining these estimates and definitively ruling out operator-specific effects.

Finally, our reliance on official administrative boundaries (municipalities) to delineate urban units, while practical for aligning CDR data with census figures, may not capture the true functional boundaries of cities shaped by human mobility and interaction patterns. As recent research argues (Chen and Yeh, 2022; Dong et al., 2024; Sotomayor-Gómez and Samaniego, 2020), mobile phone data itself can be used to derive more accurate, functional urban areas, and the choice of boundary can significantly impact scaling exponents. How our results might change when using functionally-defined boundaries deserves further attention in future work.

6 Conclusion

This study provided a national-scale empirical validation of urban scaling laws using mobile phone data from Brazil’s 100 largest cities. We demonstrated that both remote communication (phone contacts) and physical mobility (trips) scale superlinearly with population size, with exponents of 1.19 and 1.25 respectively. These patterns persist across cities with profound socioeconomic inequality and diverse infrastructure, suggesting that interaction-driven agglomeration effects reflect fundamental urban mechanisms rather than context-specific factors.

Our analysis makes three main contributions. First, we established methodological approaches for correcting sampling biases in telecommunications data and extrapolating from observed users to population-level interaction metrics. Second, we introduced the concept of effective population for infrastructure analysis, demonstrating that telecommunications infrastructure scales sublinearly with active users despite appearing superlinear when measured against nominal population. Third, we extend urban scaling theory to a developing country context characterized by extreme spatial and economic heterogeneity, testing whether scaling relationships derived primarily from wealthy nations remain valid under conditions of profound inequality.

The availability of large-scale datasets has transformed urban science into rigorous empirical testing. Our findings demonstrate that urban theories can now be systematically evaluated, enabling deeper investigation into how human interactions generate the economic advantages observed in larger cities.

Appendix A: CDR dataset

The study was conducted using data provided for research purposes by a telecommunications operator. The mobile telephony database comprises 30 days of call records from the year 2013, covering the period from March 21, 2013, to April 19, 2013, totaling 3.1 billion records spanning the entire Brazilian territory. In 2013, 85% of the Brazilian urban population used mobile phones. The operator maintained an average market share of approximately 20% across cities, with variation among different urban areas.

The dataset contains only outgoing calls made by users of the telecommunications operator. Consequently, it does not include additional traffic information, such as incoming calls from other operators, text messages (SMS), or internet traffic. Call location information was inferred from the geographic coordinates (latitude and longitude) of the antennas that processed the calls. Because the database contains only records of outgoing calls, it is not possible to infer the location of users who received calls.

As shown in Table 2, a Call Detail Record (CDR) contains information related to the date, time, and duration of calls, as well as details about the area code (Direct Distance Dialing – DDD) and the phone numbers of both the caller and the recipient. To preserve user privacy, all phone number information is encrypted. Additionally, each CDR contains details about the type of traffic used in the call (such as international roaming) and the names of both the originating and destination operators.

Table 2 CDR data structure and field descriptions

Field	Description
day	The day of the record
time	The time of the record
duration	The duration of the call
ddd_orig	The area code of the originating station
num_orig	The encrypted ID of the originating user
ddd_dest	The area code of the destination station
num_dest	The encrypted ID of the destination user
cell_id_lat	Latitude coordinate of the antenna that processed the call
cell_id_long	Longitude coordinate of the antenna that processed the call
tp_traffic	Type of call (local, international, roaming, etc.)
hold_orig	Name of the operator that processed the call at the origin
hold_dest	Name of the operator that handled the call at the destination

Description of fields contained in a single call record (CDR)

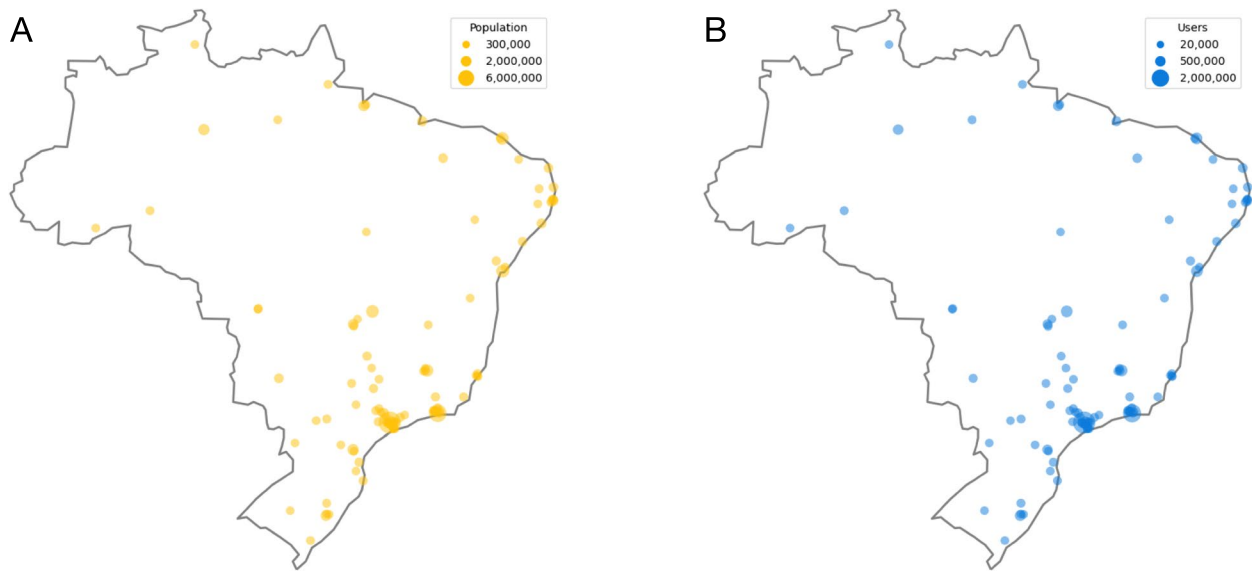


Fig. 4 The 100 most populous municipalities in Brazil analyzed in this study. **A** Node size represents the population of these 100 cities, whereas in **B**, node size represents the number of users. While they appear visually correlated, the number of users (U) grows superlinearly with population (N) (Fig. 2, left panel). Map created using data from Natural Earth (<https://www.naturalearthdata.com>), which is in the public domain, and visualized using Python/geopandas

Appendix B: The 100 most populous Brazilian municipalities

Brazil comprises 5,570 municipalities, ranging from large megacities with over 10 million inhabitants to small towns with only a few thousand residents. This diversity in urban form, climate, and culture makes Brazil an ideal setting for studying mobility and social interactions. However, analyzing all municipalities could reduce the reliability of scaling estimates, particularly for smaller cities with limited data. Therefore, this study focuses on the 100 most populous municipalities, which collectively span a wide range of climates, cultures, and economic conditions.

Population data for these cities were obtained from the 2010 IBGE census, ensuring consistency with the CDR dataset. By focusing on these 100 cities, the study balances diversity with statistical robustness. The geographic distribution of the 100 most populous municipalities analyzed in this study is shown in Fig. 4.

Appendix C: Scaling analysis

The scaling analysis conducted in this paper examines how a given metric, denoted as Y , scales with a reference metric, such as population size or the number of users. Empirical evidence (Bettencourt et al., 2007) suggests that a power-law relationship of the form

$$Y \sim N^\beta, \quad (15)$$

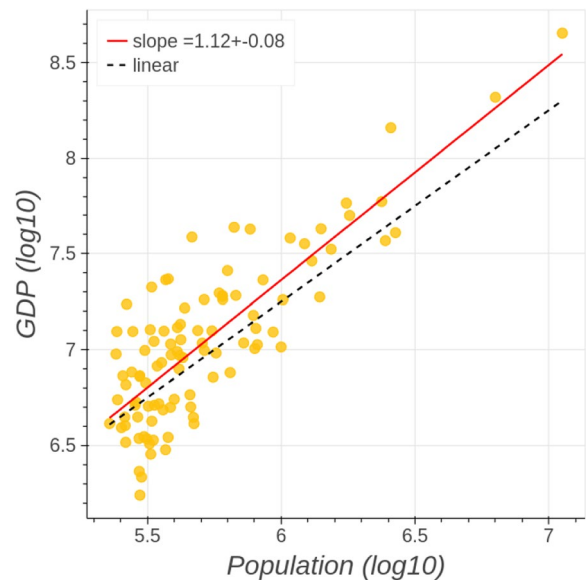


Fig. 5 Superlinear scaling of GDP with population size. Scaling analysis of GDP and estimation of the scaling exponent. The results reveal a superlinear relationship between GDP and city population size, characterized by a scaling exponent $\beta_G > 1$. This indicates that larger cities tend to generate disproportionately higher economic output per capita. The plot, presented on a log-log scale, illustrates this scaling behavior and serves as the basis for estimating the exponents reported in Table 1

holds when city population size (N) is used as the scaling parameter. In this study, we assume that a similar power-law relationship also applies when the number of users (U) is considered as the scaling parameter.

In Eq. (15), β is the scaling exponent we aim to estimate. To obtain this estimate, we first take the logarithm of both sides of Eq. (15) and then apply ordinary least squares (OLS) regression to estimate the exponent. The estimated scaling exponents for various urban metrics are presented in Table 1. Additionally, Figs. 2, 3, and 5 illustrate the scaling behavior of selected urban variables.

Appendix D: Average degree $\langle k^s \rangle$

The superlinear scaling of the cumulative degree K^s —defined as the total number of contacts among users within a specific city—is a direct consequence of the higher average degree $\langle k^s \rangle$ observed in larger cities, as shown in Fig. 6, panels (a) and (b). This indicates that individuals living in larger urban areas tend to maintain a greater number of mobile phone contacts on average.

In the present work, we estimate the average degree using a sample of size U drawn from the entire city

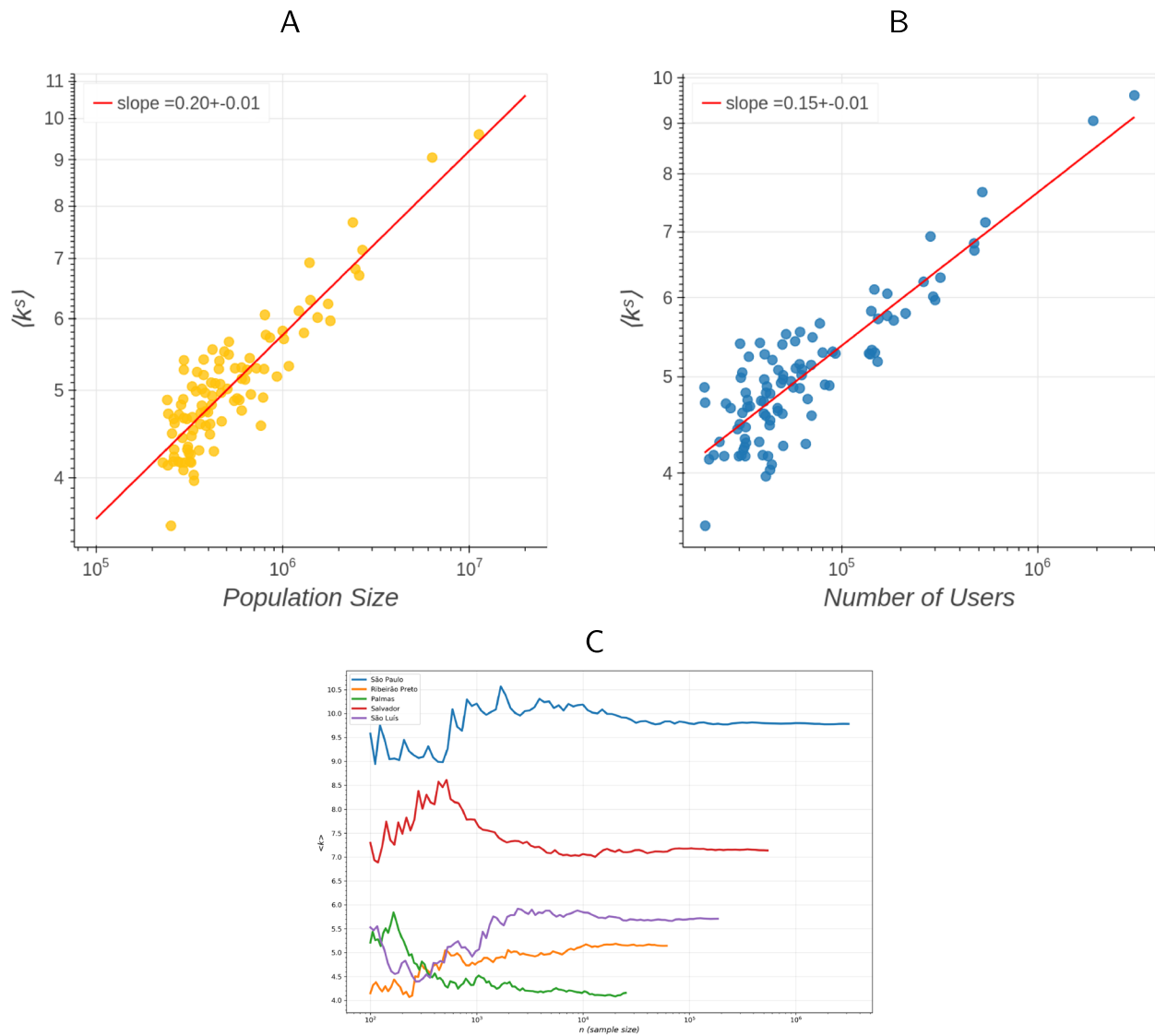


Fig. 6 Scaling and convergence of average degree. Scaling analysis of the average degree $\langle k^s \rangle$. Panel **A** presents the scaling of $\langle k^s \rangle$ with respect to city population size N , while Panel **B** shows the same analysis relative to the number of users U . Regardless of the chosen scaling metric, a clear power-law relationship emerges. The positive slope in both cases indicates that the average number of contacts per user increases systematically with city size. Panel **C** demonstrates that for cities of different sizes, regardless of the subset size n sampled from the total number of users U (with $n \leq U$), the average degree $\langle k^s \rangle$ converges as n becomes sufficiently large. This empirical convergence supports the assumption formalized in Eq. (6)

population N , where $U < N$. This naturally raises the question of whether such a sample is representative of the entire population. To address this concern, we analyze smaller subsets of the sample—subsamples of size n from the original sample of size U , with $n \leq U$. As shown in Fig. 6c, the average degree $\langle k^s \rangle$ converges as n becomes sufficiently large. This empirical observation is a manifestation of the *law of large numbers* and provides justification for the assumption expressed in Eq. (6), namely that $\langle k_i^s \rangle \approx \langle k_i \rangle$. In other words, the average degree computed from the observed users (the sample) serves as a reliable proxy for the true average degree across the entire population, which is not directly observable.

It is worth noting that for very small cities, reaching a sufficiently large sample size to ensure convergence may not always be feasible. However, this limitation does not affect our analysis, as we focus on the 100 largest Brazilian cities, which generally have sufficient numbers of users to support the stability of this

assumption. For example, consider the case of Palmas (green curve in Fig. 6c), one of the smallest cities among those analyzed. Even in this case, the average degree exhibits convergence behavior, reinforcing the robustness of our approach.

Appendix E: Trip detection algorithm

The methodology for estimating the OD matrix, considered for calculating trips in each municipality, counts trips without distinguishing whether the user departed from home or not. The algorithm described below presents the steps used to estimate the OD matrices from the CDR data used in this study. The calculation only considers users with an identified presumed address. Trip detections are then performed, where it is verified whether two consecutive calls from the same ID are made. By ordering ID, DAY, TIME, it is observed whether there was a change in the geographic position (latitude and/or longitude) of the antenna that processed the next call.

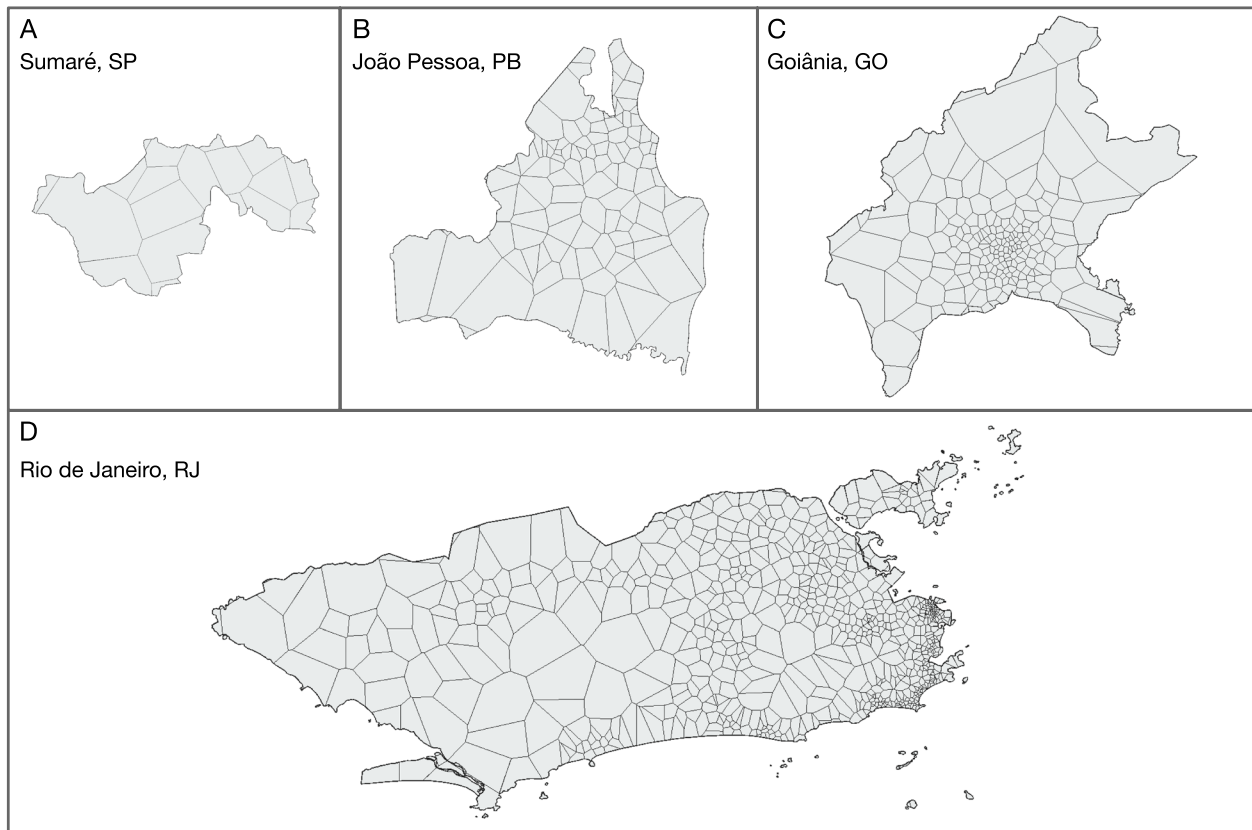


Fig. 7 Voronoi tessellation of cellular network coverage in four Brazilian cities. **A** Sumaré, SP (population ≈ 241 K). **B** João Pessoa, PB (population ≈ 723 K). **C** Goiânia, GO (population ≈ 1.3 M). **D** Rio de Janeiro, RJ (population ≈ 6.3 M). Despite antenna placement prioritizing population density and socioeconomic factors, these visualizations reveal significant heterogeneity in spatial coverage across cities of varying sizes. Map created using data from IBGE (<https://www.ibge.gov.br/geociencias/organizacao-do-territorio/malhas-territoriais/15774-malhas.html>), which is open source. The map view and Voronoi tessellation were created using QGIS open-source software

The main idea of trip estimation in this work is to capture motorized trips. A trip is considered a trip if a minimum of 2 km was “traveled” in a minimum of 30 minutes and a maximum of 4 hours. The distance “traveled” between consecutive calls is the linear distance between the telephone towers that processed the calls. In other words, it is an approximate distance, taking into account that the user makes calls within the antenna coverage area and that the trip route is different from a straight line. After obtaining the OD matrices for each municipality, the number of trips is obtained from the sum of all trips.

Algorithm 1 OD matrix estimation from CDR data

Input: CDR logs, the constants ΔT and L_{min} , table of distances between geographic units and $T_{min} T_{max}$
Output: OD Matrix

```

1: Begin
2:  $OD = \text{NULL}$ 
3: For each trip detected by two successive records of the same user, within distance,  $l_{ij}$ , and time interval,  $\Delta T$ 
4:   If  $l_{ij} > L_{min}$  and  $T_{min} \leq \Delta T \leq T_{max}$ 
5:      $OD_{ij} \leftarrow OD_{ij} + e_i$ 
6:   End If
7: End For
8: Return OD Matrix
9: End

```

Appendix F: Voronoi cells

Each call in the CDR dataset is associated with a specific antenna, from which a Voronoi tessellation is constructed to approximate its coverage area. Figure 7 illustrates this tessellation for four cities of different sizes. Due to varying antenna densities, these Voronoi polygons can differ significantly in size, leading to decreased positional accuracy in sparsely covered regions. Consequently, while these partitions provide a practical basis for spatial analysis, potential bias may exist in areas with fewer antennas.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43762-026-00252-x>.

Supplementary Material 1.

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Declarations

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Authors' contributions

Conceptualization: all authors; Data curation: HS, AE, RA; Formal analysis: all authors; Funding acquisition: FR, HS, AE, RM; Investigation: all authors; Methodology: all authors; Project administration: all authors; Validation: all authors; Writing – original draft: all authors; Writing – review & editing: all authors.

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Data availability

Supplementary Material is provided as a spreadsheet containing the aggregated data for the 100 largest cities in Brazil used in the present study: <https://doi.org/10.5281/zenodo.15359949>.

Declarations

Competing interests

The authors declare that they have no competing interests.

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